

Advanced (Habanero)

- Q1.** What is the primary purpose of a stem-and-leaf plot?
- (A) To display the mean of a dataset
(B) To show the distribution of a dataset
(C) To calculate the median of a dataset
(D) To visualize the range of a dataset
(E) To compare multiple datasets
- Q2.** Which of the following is a benefit of using a stem-and-leaf plot?
- (A) It can only be used with small datasets
(B) It is difficult to read and interpret
(C) It can help identify outliers in a dataset
(D) It is only useful for displaying categorical data
(E) It is not suitable for large datasets
- Q3.** What is the stem in a stem-and-leaf plot?
- (A) The leaf values
(B) The stem values
(C) The median of the dataset
(D) The range of the dataset
(E) The first digit of each value
- Q4.** How do you determine the stem values in a stem-and-leaf plot?
- (A) By finding the mean of the dataset
(B) By finding the median of the dataset
(C) By grouping the first digit(s) of each value
(D) By grouping the last digit(s) of each value
(E) By randomly selecting values
- Q5.** What is the leaf in a stem-and-leaf plot?
- (A) The stem values
(B) The leaf values
(C) The median of the dataset
(D) The range of the dataset
(E) The first digit of each value
- Q6.** Which of the following datasets would be most suitable for a stem-and-leaf plot?
- (A) A dataset with a small range of values
(B) A dataset with a large range of values
(C) A dataset with only categorical values
(D) A dataset with only numerical values
(E) A dataset with a mix of numerical and categorical values
- Q7.** How do you interpret the distribution of a dataset from a stem-and-leaf plot?
- (A) By looking at the mean of the dataset
(B) By looking at the median of the dataset
(C) By examining the shape of the plot
(D) By examining the range of the dataset
(E) By comparing it to another dataset
- Q8.** What can a stem-and-leaf plot help you identify in a dataset?
- (A) The mean and median
(B) The range and mode
(C) Outliers and skewness
(D) The standard deviation
(E) The variance

Open Ended Questions (FRQ)

- Q9.** Construct a stem-and-leaf plot for the following dataset: {12, 15, 18, 20, 22, 25, 28, 30, 32, 35}. Describe the distribution of the dataset.
- Q10.** Explain how a stem-and-leaf plot can be used to compare the distributions of two different datasets. Provide an example using the following datasets: {10, 12, 15, 18, 20} and {8, 10, 12, 15, 18}. Use *stem – and – leafplots* to support your answer.

Chef's Tips

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- Tip 1: Encourage students to use the stem-and-leaf plot to identify outliers and skewness in the dataset.
- Tip 2: Remind students to examine the shape of the stem-and-leaf plot to interpret the distribution of the dataset.
- Tip 3: Suggest that students use the stem-and-leaf plot to compare the distributions of multiple datasets.